

# IS-005 Sentinel Knowledge Graph

## Implementation Specification IS-005 - Sentinel Knowledge Graph

Status: Approved for Design

Priority: Critical

Package ID: PKG-005

Depends on: IS-001, IS-002, IS-003, IS-004

Referenced Blueprint Volumes: 14, 16, 31, 39

### Vision

The Knowledge Graph is the memory of Project Sentinel. Databases store records. The Knowledge Graph stores relationships and meaning.

### Core Principle

Knowledge is built from verified evidence. No node enters the graph without traceable evidence.

### Graph Components

Objects -> Relationships -> Evidence Links -> Knowledge Links -> Context -> History

### Node Types

Device, User, Identity, Organization, Workspace, Process, File, Certificate, Network Connection, Application, Vulnerability, Threat, Mission, Finding, Evidence, Policy, Rule, Report, Cloud Resource.

### Relationship Types

User owns Device.

Device runs Process.

Process connects\_to IP Address.

Finding supported\_by Evidence.

Policy protects Asset.

### Graph Rules

Every relationship includes Source Node, Target Node, Relationship Type, Created Time, Confidence, Evidence Reference and Version.

## **Time Awareness**

The graph supports current view, historical view and comparison view.

## **Requirements**

REQ-KNOWLEDGE-001: Every knowledge node shall reference an object.

REQ-KNOWLEDGE-002: Every relationship shall reference supporting evidence where applicable.

REQ-KNOWLEDGE-003: Knowledge shall preserve historical changes.

REQ-KNOWLEDGE-004: Queries shall respect authorization boundaries.

REQ-KNOWLEDGE-005: Confidence shall be calculated from evidence.

REQ-KNOWLEDGE-006: The graph shall support historical and current views.

REQ-KNOWLEDGE-007: The graph shall expose APIs for search and investigation.